IEEE INGARSS 2026 YP HACKATHON · NISAR L2 GCOV · BIHAR

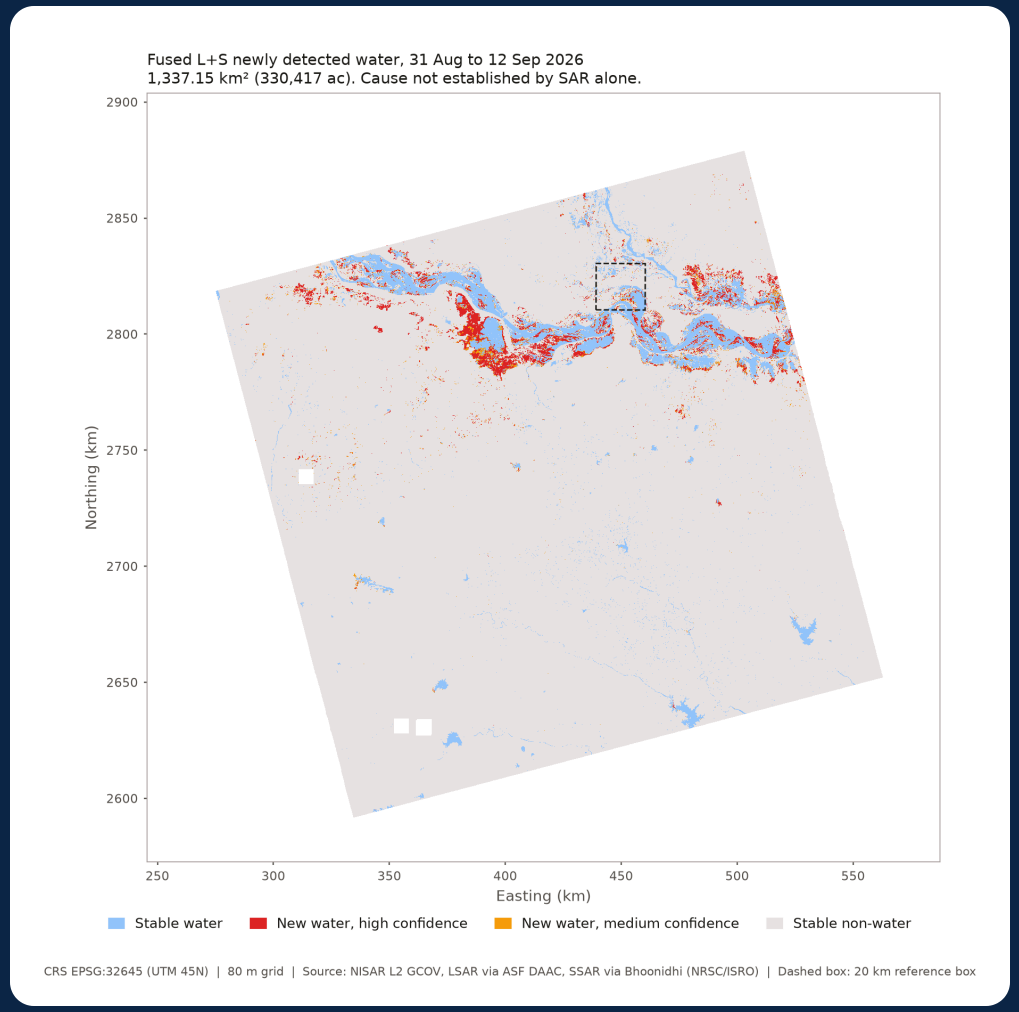
Does combining NISAR L-band and S-band give a better water map than one band alone?

- Full scene, track 98, frame 14, not just the 20 km box. Common L/S grid: 59,083.56 km² (14,599,867 acres)
- Three maps on the same 80 m grid and the same dates: L-band only, S-band only, L+S
- Checked against Sentinel-2 wherever there was no cloud

L-BAND	S-BAND	L+S
0.859	0.891	0.891

F1 against Sentinel-2 water, 12 Sep 2026, cloud-free pixels. Agreement with optical labels, not ground truth.

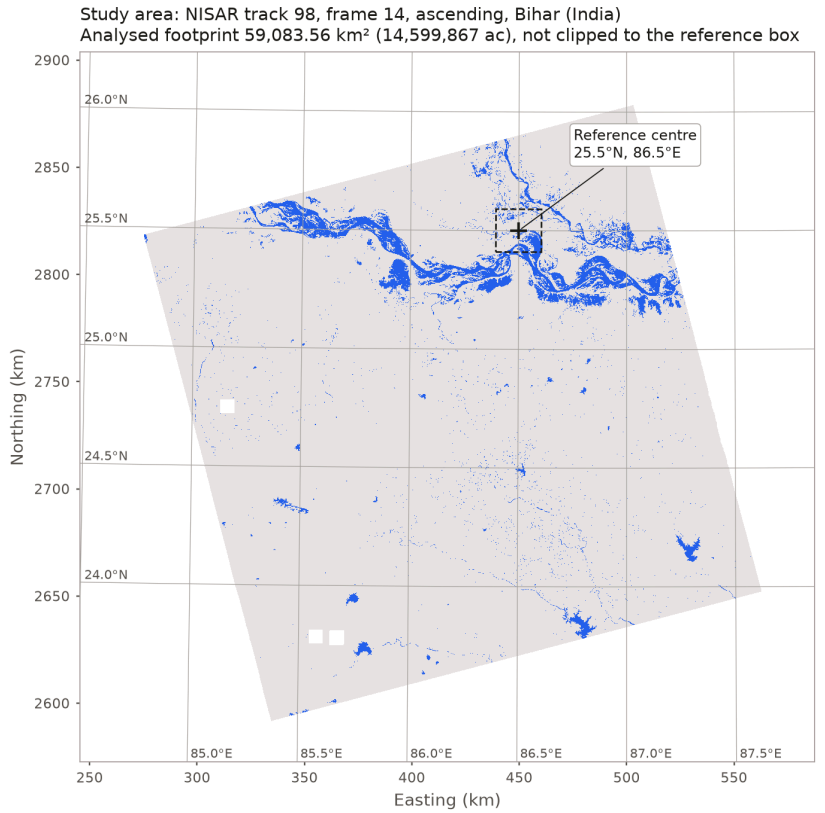
Answer: L+S beat L-band in all 3 Sentinel-2 tests. It did **not** beat S-band: equal on 12 Sep, lower on 31 Aug and for new water.



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STUDY AREA AND DATA

We processed the full NISAR scene; the 20 km box was only for finding it



Dates: L-band 26 Jul, 31 Aug, 12 Sep, 18 Oct 2026; S-band 31 Aug, 12 Sep 2026. Blue: stable water 31 Aug and 12 Sep (L+S).
White squares inside the footprint: 6.5 km cut around small no-data spots (same 40-pixel rule as the swath edge).

Analysed footprint | Stable water (L+S) | 20 km reference box

CRS EPSG:32645 (UTM 45N) | 80 m grid | Source: NISAR L2 GCOV, LSAR via ASF DAAC, SSAR via Bhoonidhi (NRSC/ISRO) | Dashed box: 20 km reference box

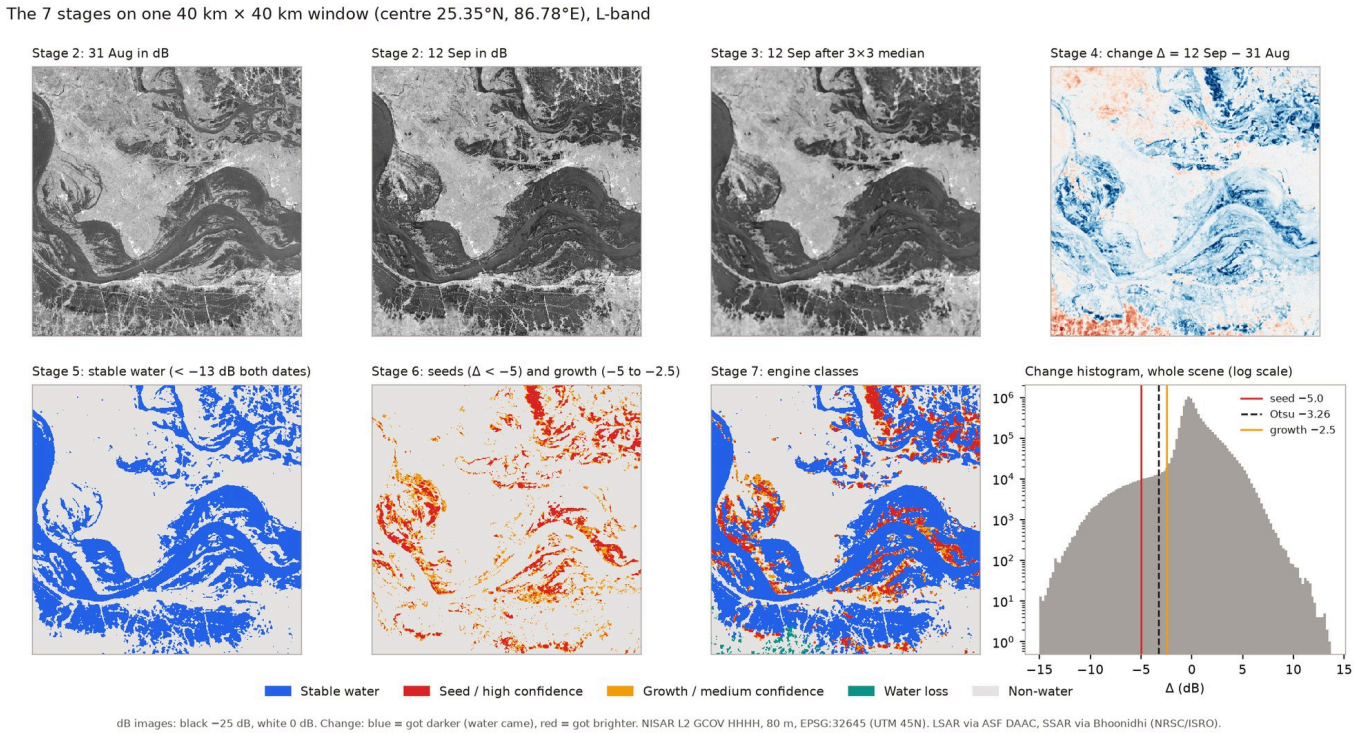
2026	L-BAND 1.239 GHZ	S-BAND 3.2 GHZ
26 Jul	DHDH	not used
31 Aug	DHDH	DHNA
12 Sep	DHDH	DHNA
18 Oct	DHDH	not used

- L-band from ASF DAAC, S-band from Bhoonidhi (NRSC/ISRO). Ascending track 98, frame 14, 10 m grid, EPSG:32645
- L and S were taken seconds apart on 31 Aug and 12 Sep, so the three-way test uses that pair
- HHHH for detection, HVHV for fusion and polarization
- Sentinel-2 Level-2A is used only as independent reference

Catalogue search: ASF lists L-band full-pol on descending track 120, frame 76 (e.g. 2 and 14 Sep), not processed here. Bhoonidhi showed no S-band compact-pol for the dates searched. HHHV (S-band) is not an intensity, so it is not used as one.

METHOD, ONE BAND

One 7-stage pipeline per band; only the thresholds change between L and S



The 7 stages on one 40 km window, L-band, 31 Aug → 12 Sep 2026.

1. **Check the pair:** same term (HHHH) and product (GCOV) on both dates
2. **Decibels:** $\sigma^o = 10 \cdot \log_{10}(\bar{P})$; below −40 dB is no data; cut 40 pixels (3.2 km) at every no-data edge
3. **Speckle:** median of the valid 3×3 neighbours
4. **Change:** Δ = dB(after) − dB(before)
5. **Stable water:** below the water level on both dates
6. **New water:** seeds below the seed level, grown into 8 neighbours below the growth level; groups under 8 pixels removed. Otsu is reported only: T_seed = min(−5.0, T_Otsu, −3.0) = −5.0 dB
7. **Classes** 0, 1, 2, 3, 5 and areas. Class 4 needs an InSAR coherence layer, not used here

ONE LAYER
1.16 billion px
4.63 GB if read at once

WE KEEP
72 MB
1,024-row strips, 8×8 average to 80 m

EACH PAIR
44–65 s
1.87–2.32 GB peak, 4-core machine

THRESHOLD (DB)	L	S
Stable water, both dates	−13.0	−12.5
Seed change	−5.0	−4.8
Growth change	−2.5	−2.2
Otsu (reported only)	−3.26	−3.47

METHOD, L+S FUSION

Fusion is four if/then rules on one shared 80 m grid

1. Sandbar rule — REMOVES FALSE WATER

L HH < −13 dB but S HH ≥ −10.5 dB on either date → not stable water

A surface can look smooth to the long L wave and rough to the short S wave.

2. Wind rule — ADDS WATER

L HH < −15.5 dB and HV < −22.0 dB on both dates → stable water, even if S missed it

3. Canopy rule — ADDS CANDIDATES

L HH drops ≥ 3.26 dB, later HV < −20 dB, later HH < −12 dB → new-water candidate

Not verified: optical data cannot see under plants.

4. S-band gate — MAIN FILTER

Every new-water candidate needs S HH < −10.5 dB on 12 Sep

Rule names describe intent, not proven surface types. Both grids and their 720 m / 2,880 m offset are exact multiples of 80 m, so L and S pixels line up with no resampling.

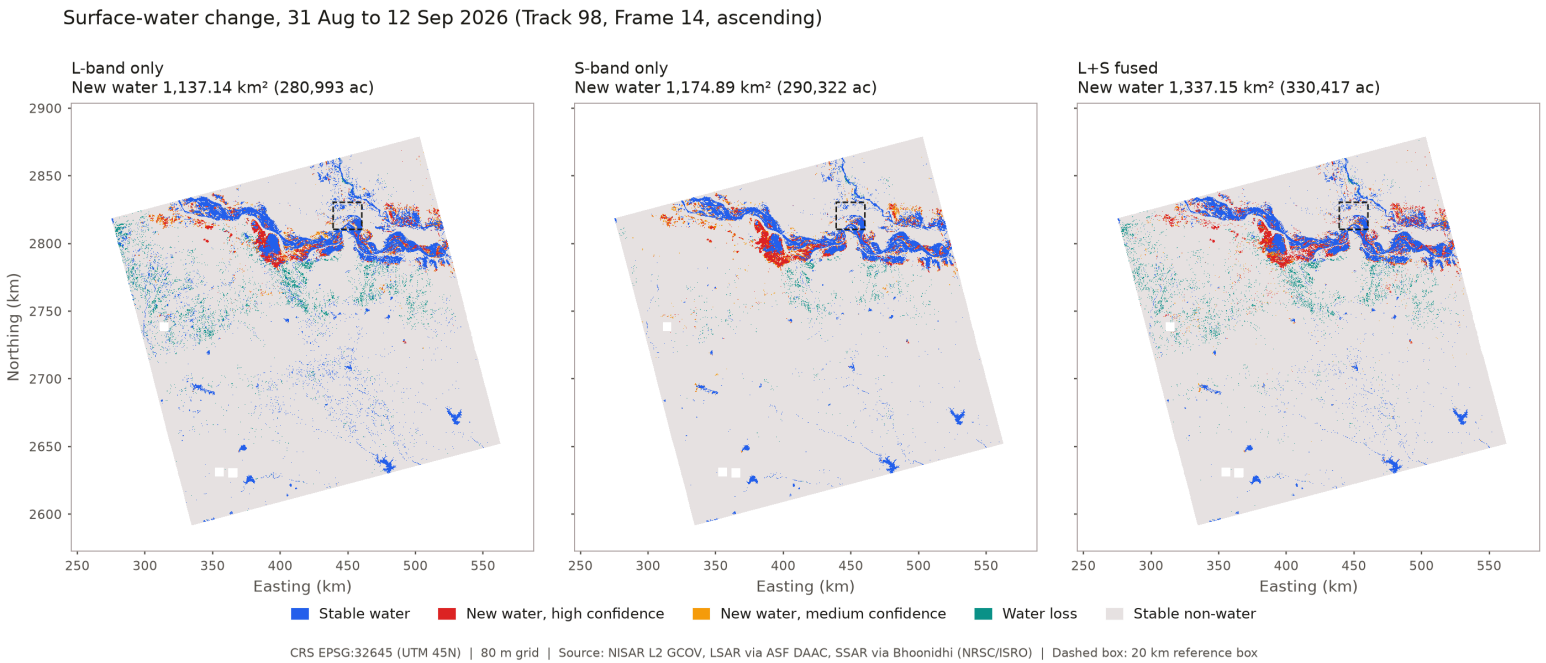
WHAT EACH RULE DOES: SWITCH IT OFF AND RUN AGAIN

RUN	STABLE WATER	NEW WATER
All rules	2,652.29 km² 655,395 acres	1,337.15 km² 330,417 acres
No sandbar rule	2,684.26 km² 663,296 acres	1,322.69 km² 326,845 acres
No wind rule	2,381.78 km² 588,551 acres	1,386.80 km² 342,685 acres
No canopy rule	2,652.29 km² 655,395 acres	1,113.61 km² 275,180 acres
No S-band gate	2,652.29 km² 655,395 acres	1,675.85 km² 414,111 acres

The **S-band gate** is the strongest control: it removed 338.70 km² (83,695 acres) of candidates. Moving it by ±1 dB moves new water between 1,208.44 and 1,449.29 km² (298,611 to 358,128 acres).

RESULTS, 31 AUG → 12 SEP 2026

The bands agree on the main pattern; L-band marks much more area as water



NEW WATER ON THE COMMON FOOTPRINT

L-BAND
1,137.14 km²
280,993 acres

S-BAND
1,174.89 km²
290,322 acres

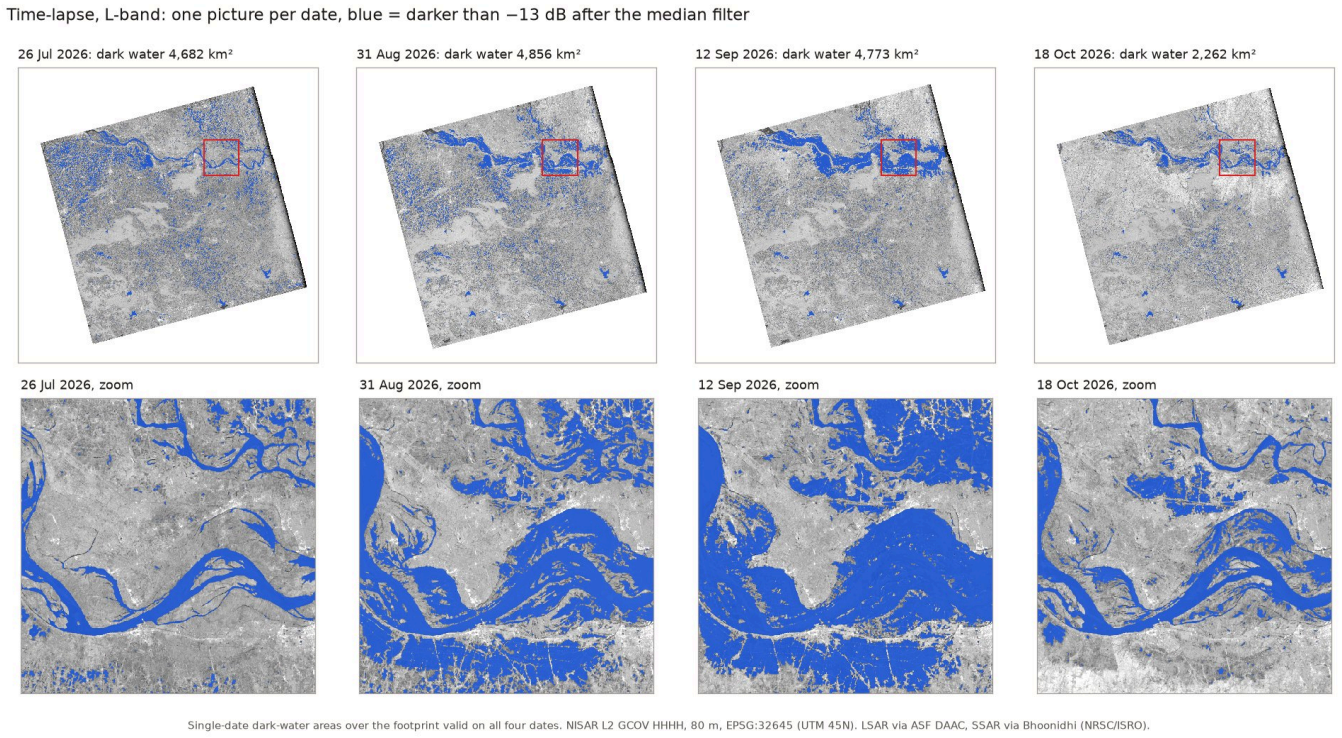
L+S
1,337.15 km²
330,417 acres

L and S new water overlap by 858.56 km² (212,155 acres), Jaccard 0.59. L+S is partly larger by design, because its candidates come from both bands.

SAME FOOTPRINT	STABLE WATER	WATER LOSS	WATER ON 12 SEP
L-band	3,314.83 km² (819,112 acres)	888.84 km² (219,637 acres)	4,451.97 km² (1,100,105 acres)
S-band	2,546.37 km² (629,222 acres)	214.98 km² (53,123 acres)	3,721.26 km² (919,544 acres)
L+S	2,652.29 km² (655,395 acres)	723.37 km² (178,749 acres)	3,989.43 km² (985,811 acres)

CHANGE DETECTION, L-BAND, FOUR DATES IN 2026

Water spread along the river into September; about half the dark water was gone by mid-October



Single-date dark-water areas over the footprint valid on all four dates. NISAR L2 GCOV HHHH, 80 m, EPSG:32645 (UTM 45N). LSAR via ASF DAAC, SSAR via Bhoonidhi (NRSC/ISRO).

DARK WATER (BELOW −13 DB), 40 KM RIVER WINDOW

26 JUL	31 AUG	12 SEP	18 OCT
213 km²	**617 km²**	**812 km²**	**299 km²**
52,609 acres	152,400 acres	200,608 acres	73,872 acres

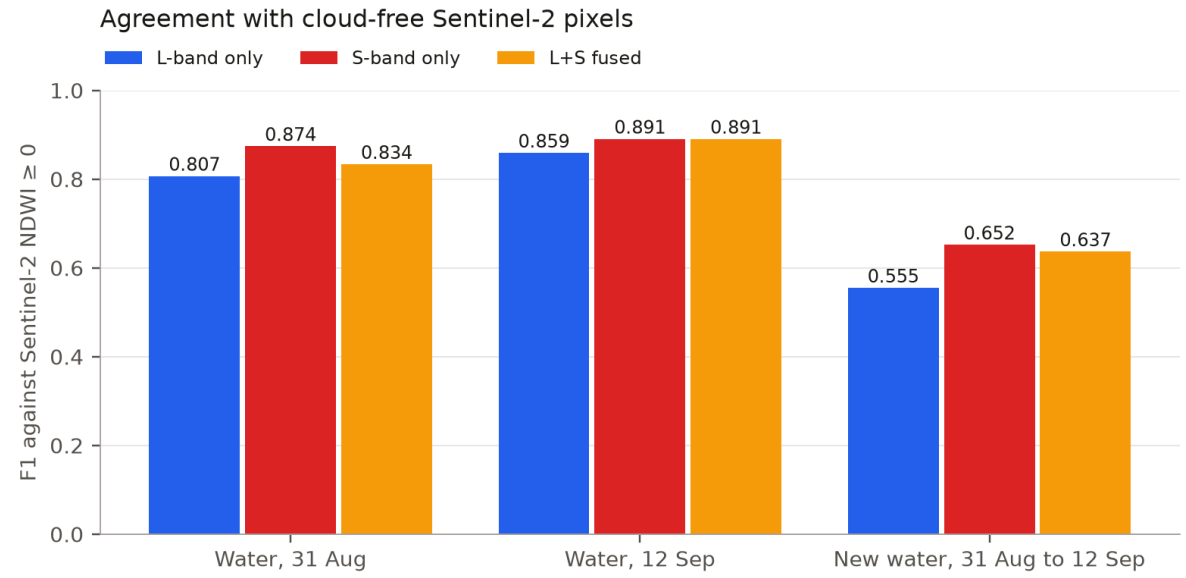
L-BAND PAIR	NEW WATER	WATER LOSS
26 Jul → 31 Aug	1,695.81 km² 419,043 acres	1,471.24 km² 363,553 acres
31 Aug → 12 Sep	1,166.21 km² 288,177 acres	889.96 km² 219,914 acres
12 Sep → 18 Oct	239.60 km² 59,207 acres	2,551.94 km² 630,597 acres

Each pair on its own valid area.

1,282.78 km² (316,982 acres) of L+S new water lies outside any water seen on 26 Jul or 31 Aug: the problem statement's "potential temporary surface inundation". SAR alone does not tell us why it appeared.

VALIDATION AGAINST SENTINEL-2 (NDWI ≥ 0, FIXED BEFORE SCORING, CLOUD-FREE PIXELS ONLY)

L+S beats L-band, mainly through fewer false alarms, but not S-band



Optical labels cannot see water under vegetation. Agreement, not ground truth.

12 SEP (S-2 ABOUT 5 H LATER)	PRECISION	RECALL	F1
L-band	0.826	0.895	**0.859**
S-band	0.911	0.872	**0.891**
L+S	0.900	0.881	**0.891**

Scored 7,946.03 km² (1,963,507 acres), about 14% of the footprint, because of cloud.

F1	L	S	L+S
Water, 31 Aug (S-2 29 Aug)	0.807	0.874	0.834
New water, 31 Aug → 12 Sep	0.555	0.652	0.637
Water, 18 Oct (L only)	0.744	–	–

Agreement with optical labels, not ground truth. Confusion matrices: `VALIDATION.md`.

Checking the sandbar rule

Pixels dark in L but bright in S were water only **8.7%** (12 Sep) and **9.2%** (31 Aug) of the time, against 80.3% and 70.3% where both were dark.

Not all sand: bare ground led on 31 Aug, vegetation on 12 Sep.

Polarization, L-band, 12 Sep

F1 0.848 with HH, 0.835 adding HV, 0.854 with HH + 2·HV. HV added little here.

S-band was the strongest single band here; fusion fixed most of L-band's over-mapping

LIMITS

- Cloud: only about 14% of the footprint could be scored on 12 Sep
- Optical images cannot see water under crops, so L-band's possible advantage there is untested
- Thresholds are hand-set; the S-band gate is the main sensitivity
- New water is a backscatter drop; features narrower than about 240 m are removed
- L-band full-pol (track 120) was found but not processed
- No cause is assigned to any water change

CONCLUSIONS

- L+S improved on L-band in every Sentinel-2 test and gave the best new-water precision (0.734)
- L+S did not beat S-band alone in these tests
- The whole scene runs on a normal workstation in about 2 GB of memory
- Next: labels under vegetation, better thresholds, more L+S dates, full-pol data

Check it yourself. `python3 run_bihar_analysis.py` checks all file hashes, prints every number in these slides and ends with ALL CHECKS PASSED.